Politecnico di Milano

Data and Information Quality - 056490

Data Preparation Pipeline

on the dataset

Comune-di-Milano-Servizi-alla-persona-parrucchieri-estetisti.csv

Project ID: 42
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Authors

Alessio Tagab
Email:
alessio.tagab@mail.polimi.it
Matricola: 270485

Matteo Trombetta
Email:
matteo2.trombetta@mail.polimi.it
Matricola: 279729

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Chapter 1

Data Profiling and Quality Assessment

1.1 Initial Dataset Overview

The project began with a general assessment of the raw dataset, which initially contained approximately 3,900 rows describing personal service businesses in Milan. The primary goal of this phase was to understand the structural integrity of the data and identify quality dimensions that required intervention. We also performed a first brief deduplication step to check and clean possible exact equal duplicates directly from the dataset. For the data quality assessment we can just assess the completeness and the duplication, we do not have enough information also for the consistency.

1.2 Data Quality Dimensions

Initial inspection revealed significant issues regarding atomic integrity and completeness. The dataset was heavily denormalized; specifically, the Ubicazione column contained a concatenation of street names, street numbers (Civico), and administrative zones (ZD), making it impossible to perform a correct analysis without parsing. Furthermore, the Tipo esercizio pa column, intended to categorize business activities, had a very heterogeneous structure, combining unstandardized and undefined administrative codes (e.g., "TIPO A", "TIPO B") with natural language descriptions.

1.3 Profiling Tools

Data Profiling was performed with the use of the y_data_profiling library, which provided both an html and a json file rendering the full picture of the initial situation of this dirty dataset. The same library was use to obtain the respective files once the dataset was eventually clean.

1.4 Initial Issues Identified

A statistical analysis of the numerical columns highlighted completeness issues in the Superficie altri usi column, which was missing in the vast majority of records, rendering it statistically insignificant. Also, the Superficie lavorativa column, while mostly complete, exhibited high variance, indicating a need for outlier detection. Finally, a preliminary check for duplicates suggested that while exact row duplication was minimal, the dataset likely contained duplicates (records representing the same entity but differing slightly in spelling or formatting).

	Tipo esercizio pa	Ubicazione	Tipo via	Via	Civico	Codice via	ZD	Prevalente	Superficie altri usi	Superficie lavorativa
0	NaN	LGO DEI GELSOMINI N. 10 (z.d. 6)	LGO	DEI GELSOMINI	10	5394.0	6	NaN	NaN	55.0
1	NaN	PZA FIDIA N. 3 (z.d. 9)	PZA	FIDIA	3	1144.0	9	CENTRO MASSAGGI RILASSANTI NON ESTETICI	2.0	28.0
2	NaN	VIA ADIGE N. 10 (z.d. 5)	VIA	ADIGE	10	4216.0	5	CENTRO BENESSERE	2.0	27.0
3	NaN	VIA BARACCHINI FLAVIO N. 9 (z.d. 1)	VIA	BARACCHINI FLAVIO	9	356.0	1	TRUCCO SEMIPERMANENTE	NaN	NaN
4	NaN	VIA BERGAMO N. 12 (z.d. 4)	VIA	BERGAMO	12	3189.0	4	NaN	NaN	50.0
5	NaN	VIA BOTTEGO VITTORIO N. 13 (z.d. 2)	VIA	BOTTEGO VITTORIO	13	2521.0	2	CENTRO BENESSERE	5.0	90.0

Figure 1.1: First view of the dataset before the data cleaning pipeline

Chapter 2

Data Cleaning

2.1 Data Transformation and Standardization

The first phase of cleaning focused on standardizing formats and parsing unstructured text into atomic, usable columns. This was a critical step to ensure that following deduplication and analysis would be accurate. To do this we went through a column-by-column approach, considering each column and its characteristic one by one. The general strategy was to try to keep as much information and data as possible without removing too many records.

2.1.1 Address Parsing and Atomization

The most complex transformation involved the Ubicazione column. A sophisticated Regular Expression (Regex) strategy was implemented to parse this string and extract four distinct components: the street name (Indirizzo), the street number (Civico), the administrative zone (ZD), and the street code (Codice via). This extraction process prioritized capturing the details embedded in the text strings, such as separating "VIA ADIGE N. 10" into its constituent parts.

2.1.2 Activity Taxonomy Standardization

At the same time, the business activity classifications were normalized. The raw Tipo esercizio pa and Prevalente columns contained a mix of codes and descriptions. We decided to re-elaborate these two columns: we removed them and created two clearer ones, Attività principale and Tipo di Attività. This decomposition of the column is needed for a deeper comparison.

In the column Attività principale we decided to insert a new standardized classification that suited better the types of business of our dataset. We did this by

defining a mapping called tipo_mapping, this was used to standardize all the information we were able to extract from the original columns that could provide a classification for each record. To do this we made use of a dummy column to extract information. The final classification we defined for this column is the following: this was built according to the information already present in the dataset (TIPO A-B-C-D) and was updated and integrated with additional ones (TIPO E-F).

Code	**Description**
TIPO A	Estetica Manuale
TIPO B	Centro Abbronzatura
TIPO C	Estetica Dimagrimento
TIPO D	Estetica App. Elettromecc.
TIPO E	Centro Benessere
TIPO F	Tatuaggi

Table 2.1: Standardized Attività Principale Classification

Then with the column Tipo di Attività we implemented a similar approach; the idea was to insert into this column a set of pre-defined strings that described which types of activity are performed by each record (to try and give a more complete description of each record). Also, here we made up a dummy column to extract the information. The mapping we decided to implement here is called activity_tipo_mapping, this ended up defining a set of activity types as following:

- ACCONCIATORE
- PARRUCCHIERE
- SOLARIUM
- CURA DELLA PERSONA
- TRUCCATORE
- MASSAGGI
- TATUAGGI
- DIMAGRIMENTO
- TRATTAMENTI ELETTROMECCANICI

For address-related attributes (Via, Tipo via, Civico, ZD), a hierarchical priority logic was established. Where original fields were null, they were systematically populated with the corresponding extracted values. In cases of conflict—where the original value differed from the extracted one—the extracted components derived from Ubicazione were prioritized as the "source of truth," under the assumption that the full address string contained the most current and detailed information. A similar reconciliation process was applied to business activities. Missing values

in the Attività principale column were imputed using the parsed results from the standardization step. For Activity Types, a union approach was adopted: services identified in the free-text description that were missing from the structured column were appended to the existing data. This ensured a comprehensive listing of all services offered by each establishment. To preserve data readability and minimize sparsity, the 'Tipo di Attività' field has been kept as comma-separated strings rather than normalized into separate columns. This approach ensures clarity for manual review while avoiding the inefficiency and high number of zeros associated with sparse matrix representations.

This phase concluded with the removal of temporary extraction columns, resulting in a single, unified set of high-quality attributes.

2.2 Error Detection and Correction

Once the data was structurally fine, the focus shifted to validity and consistency, specifically addressing missing values and outliers.

2.2.1 Handling Missing Values

The approach to missing data was selective, ensuring that no valuable information was discarded unnecessarily. The Superficie altri usi column was dropped entirely due to its excessive missingness (over 90%), as imputation would have introduced significant bias. For spatial attributes like Codice via and ZD, a "spatial mode imputation" was used. By grouping records by their street name (Indirizzo), the most frequent zone or street code associated with that street was identified and used to fill missing values, leveraging the inherent spatial correlation of the data. For the Superficie lavorativa (workspace surface area), a more nuanced imputation strategy was required. Rather than filling missing values with a global average, which would distort the data, a group-mean imputation was applied. The mean surface area was calculated specifically for each Attività principale. This ensured that a missing value for a large "Centro Benessere" was imputed with a value appropriate for that category, rather than a value skewed by smaller entities like "Nail Salons."

2.2.2 Outlier Detection

The dataset was also scrubbed for validity errors. In the Civico column, values exceeding four digits (e.g., "40945") were identified as data entry errors and removed.

For the surface area, statistical methods including Interquartile Range (IQR) and Z-Score analysis were utilized to detect anomalies.

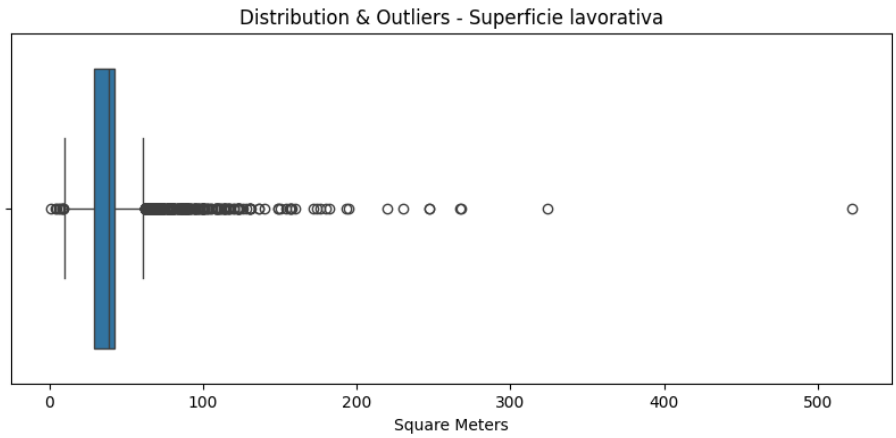

Figure 2.1: Outlier detection using the IQR method

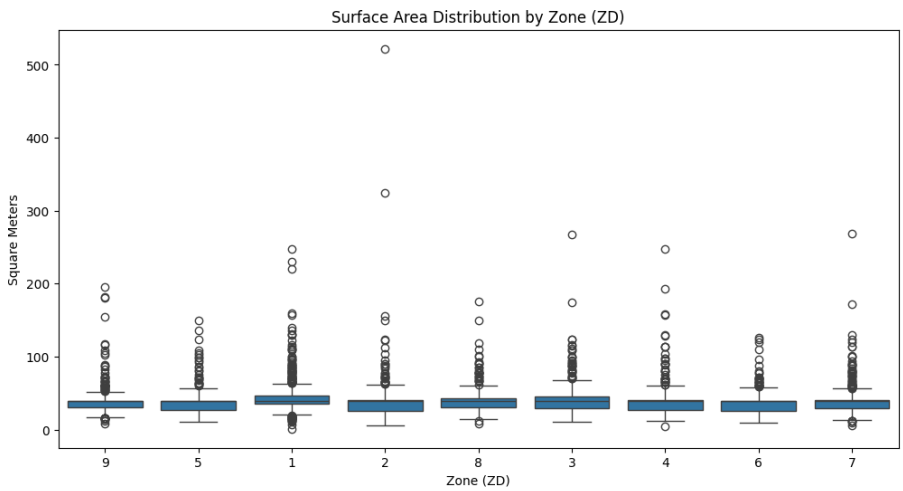

Figure 2.2: Outlier detection using the Z-Score method

While several records showed surface areas significantly larger than the mean (Z-Score > 3), a domain-driven decision was made to retain them. In the context of commercial real estate, large luxury spas or multi-purpose centers can legitimately occupy large spaces and removing them would have artificially truncated the dataset's true distribution.

2.3 Data Deduplication and Entity Resolution

The final and most complex stage of the cleaning pipeline was the deduplication process. Since the dataset contained records that referred to the same physical business

but differed slightly in specific attributes (e.g., one record listing "Parrucchiere" and another "Solarium" for the same address), a simple exact deduplication was insufficient. A probabilistic Record Linkage approach was implemented to identify and merge these entities.

2.3.1 Blocking Strategy

To make the comparison computationally feasible, the dataset was indexed using a "blocking" technique. Instead of comparing every record against every other record, comparisons were restricted to clusters of records that shared a specific "Blocking Key." The primary key used was a concatenation of Address (Indirizzo) and Street Number (Civico). This assumption held that any duplicate business must, at a minimum, be located at the same physical address.

2.3.2 Conflict Resolution Logic

Once potential duplicate pairs were identified within these blocks, they were analyzed for inconsistencies. A hierarchical resolution strategy was applied to harmonize conflicting data, ensuring the final entity contained the most accurate information available.

Spatial Consistency

The first check addressed inconsistencies in administrative data. For pairs where the Street Code (Codice via) matched but the Zone (ZD) differed, a frequency-based correction was applied. The system calculated the Mode (most frequent value) of the ZD for that specific Street Code across the entire dataset. Rows deviating from this consensus were flagged as errors and removed, ensuring that every street was assigned to its correct administrative zone. A similar logic was applied to fix inconsistencies in Codice via based on the Street Name.

Service Harmonization

A common issue was multiple records for the same salon listing partial services (e.g., Record A listed "Parrucchiere", Record B listed "Solarium"). Rather than discarding one, a "Smart Merge" function was implemented. This function performed a set union of the text fields, combining unique values into a comprehensive list (e.g., "Hairdresser, Solarium"). This ensured no service data was lost during the merge.

Principal Activity Standardization

For conflicts in the Attività principale (e.g., one record labeled "Estetica Manuale" and another "Centro Benessere"), a Majority Vote system was employed. The value that appeared most frequently for that specific location group was selected as the definitive category. This process is justified by the fact that all the activity types the salons offered are listed in the 'Tipo di Attività' column.

Numerical Outlier Resolution

Discrepancies in Superficie lavorativa (Surface Area) were resolved using a statistical distance method. For each conflicting pair, the system calculated the absolute difference between the reported surface area and the average surface area for that specific business type (e.g., the average size of all "Type A" salons). The record with the value statistically closest to the mean was retained, while the outlier was discarded. This prevented data entry errors (such as extra zeros) from skewing the final dataset.

2.3.3 Address Fuzziness and Final Cleanup

After resolving atomic attributes, a second pass of Record Linkage was performed using a looser blocking key (Civico + ZD + Codice via) to detect fuzzy duplicates in the Street Name itself. In rare comparisons, the more standard address format was manually prioritized.

	Indirizzo	Civico	Codice via	ZD	Attività principale	Tipo di Attività	Superficie lavorativa
0	PZA FIDIA	3	1144	9	CENTRO BENESSERE (E)	MASSAGGI	28.0
1	VIA ADIGE	10	4216	5	CENTRO BENESSERE (E)	MASSAGGI	27.0
2	VIA BARACCHINI FLAVIO	9	356	1	ESTETICA MANUALE (A)	TRUCCATORE	39.0
3	VIA BOTTEGO VITTORIO	13	2521	2	CENTRO BENESSERE (E)	MASSAGGI	90.0
4	VIA CUCCHIARI DOMENICO	25	7224	8	CENTRO BENESSERE (E)	MASSAGGI	176.0

Figure 2.3: Final view of the dataset after the data cleaning pipeline

Chapter 3

Conclusion and Final Assessment

The execution of this Data Quality pipeline successfully transformed a fragmented and dirty dataset into a structured, reliable resource. The final assessment confirms that the Critical Data Elements—specifically the address, business type, and size—now satisfy the dimensions of completeness, consistency, and uniqueness. The structural conflicts in the address fields have been resolved through atomization, and the lexical ambiguity in business activities has been replaced by a standardized taxonomy. Furthermore, the intelligent imputation strategies ensured that the dataset retained its statistical validity without introducing noise.